

AMAR TOPALović

Software Developer | Full-Stack Development

Sarajevo, Bosnia and Herzegovina

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[GitHub](#)

[LinkedIn](#)

PROFILE

Third-year Software Engineering student with backend engineering internship experience and hands-on full-stack development using JavaScript, TypeScript, React, Express, and MongoDB. Built applications involving REST APIs, authentication, background processing, AI integration, automated testing, and deployment. Open to full-time positions, part-time positions, and internships in Bosnia and Herzegovina, with remote, hybrid, or on-site arrangements.

TECHNICAL SKILLS

- **Main stack:** JavaScript, TypeScript, React, Node.js, Express, MongoDB.
- **Frontend:** Tailwind CSS, React Router, React Flow, responsive interfaces.
- **Backend and data:** Node.js, Express, REST APIs, Mongoose, Redis, BullMQ, authentication, role-based access control, schema validation.
- **Development tools:** Git, GitHub, Docker, GitHub Actions, OpenAPI/Swagger.
- **AI assisted development and integration:** Claude Code, Codex, Ollama, Anthropic Claude API, OpenAI API, structured model outputs.
- **Basic knowledge:** Java, Python, C++.
- **Additional experience:** PostgreSQL, MySQL, and working with Internet of Things (IoT) devices.

EXPERIENCE

Backend Engineer Intern — FlyRank AI

Remote | July–September 2026

- Applied concepts including task design, prompting, API contracts, retrieval and grounding, evaluation, and operations. Practiced taking software work through implementation, testing, documentation, and deployment.
- Developed consistent professional working habits through **34 completed practical assignments and 43 attended program meetings**, within at least **250 documented hours** of practical work, training, and learning (see [Recommendation Letter](#) on LinkedIn).
- Gained practical experience in AI-assisted software engineering, using Claude Code as the primary AI coding assistant throughout engineering assignments and AI fluency training.

- Independently designed and built a **multi-tenant lead-capture platform** using TypeScript, Express, React, MongoDB, and Redis, with embeddable widgets, a contact-management dashboard, and live analytics.
- Implemented authentication, role-based permissions, workspace data isolation, public API validation, and background email and webhook delivery with retries.
- Delivered a mentor-reviewed capstone and deployed it as a public portfolio demo using synthetic data, supported by Docker, GitHub Actions CI, API documentation, and **887 automated tests**.

Capstone: [Embeddable Widget & Lead Capture Platform](#)

PROJECTS

AI Work & Study Agent · [GitHub](#)

React, Node.js, Express, MongoDB, Tailwind CSS, Ollama, Claude API

- Built a full-stack productivity application that converts natural-language messages into study logs, notes, habits, and reflections, with goals, charts, a focus timer, and AI-generated learning reports.
- Implemented interchangeable AI providers, structured responses, and a fallback classifier. Added token revocation and input validation, and corrected goal-tracking logic.

Product Listing Enrichment API · [GitHub](#)

Node.js, Express, MongoDB, Mongoose, Zod, Ollama

- Built an asynchronous API that processes product listings into categories, summaries, data-quality flags, and confidence scores, with schema validation and a repair attempt for malformed model responses.
- Implemented atomic job claiming, crashed-worker recovery, and bounded retries with persistent backoff. Tested three concurrent workers against 40 queued jobs with no duplicate claims.

AI Decision Flow · [GitHub](#)

React, TypeScript, React Flow, Express, Inngest, OpenAI API

- Built a visual workflow editor for connecting YES/NO decision nodes and executing them through an LLM, with highlighted execution paths and continuously updated decision logs.
- Used Inngest steps to retry individual model decisions, validate workflow structure, and enforce an execution limit. Sanitized outgoing errors to reduce exposure of sensitive information.

Natours · [GitHub](#) ·

Node.js, Express, MongoDB, Mongoose, Pug, Mapbox

- Developed a tour discovery and review application with REST APIs, filtering, sorting, pagination, authentication, role-based permissions, and password recovery.
- Implemented geospatial queries, aggregation-based tour statistics, and review-rating recalculation after review creation, updates, and deletion.

The Polite Scraper — FlyRank Internship Project · [GitHub](#)

JavaScript, Node.js, Cheerio, Zod

- Built a command-line scraping pipeline that extracted and validated 60 book records from 63 pages of a practice bookstore.
- Added request delays, local caching, timeouts, selective retries, and per-page failure handling, with separate outputs for valid records, rejected records, and run diagnostics.

EDUCATION

Software Engineering — International University of Sarajevo (IUS)

Sarajevo, Bosnia and Herzegovina

2024–June 2028, expected graduation

- Currently in the third year of study.
- Academic programming and database experience includes Java and MySQL.

CERTIFICATES OF COMPLETION

- **Backend AI Engineering — FlyRank.ai**, September 2026.
- **AI Fluency — FlyRank.ai**, September 2026.

LANGUAGES

- **Bosnian:** Native.
- **English:** Fluent.